# FINITENESS AND CONTINUITY OF THE GALOIS ACTION ON THE CONNECTED COMPONENTS OF A QUASI-COMPACT BASE CHANGE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is complete as written and ends with the verification of the hypotheses in the finite type case; no step is referred to the reader.

*Editorial note.* The counted spans carry 5 navigation stubs and no bibliography citation.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

*Editorial note.* The label lemma-galois-action-connected-components-continuous is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- Connected components cover and are the maximal connected subsets. Topology, Lemma topology-lemma-connected-components, invoked at proof lines 1303-1304. Supplies the covering used for the finiteness conclusion.
- Descent of quasi-compact opens through a finitely generated subalgebra. Same chapter, Lemma lemma-descend-open, invoked at proof lines 1307-1308. Supplies that every connected component of the base change is defined over a finite subextension.
- Finite type over a field implies Noetherian, and a Noetherian space has finitely many irreducible components. Morphisms, Lemmas morphisms-lemma-base-change-finite-type and morphisms-lemma-finite-type-noetherian, and Properties, Lemma properties-lemma-Noetherian-irreducible-components, invoked at proof lines 1310-1316. Supplies that the hypotheses hold in the finite type case.

*Source reference.* Original lemma 0388: sources/stacks-varieties/pinned-original-source.tex, source lines 812–812. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 01T4: sources/stacks-chow/tags.txt, source lines 2361–2361. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

*Source reference.* Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 0BA8: sources/stacks-chow/tags.txt, source lines 14201–14201. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 004T: sources/stacks-chow/tags.txt, source lines 185–185. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $k$ be a field, with separable algebraic closure $\overline{k}$. Let $X$ be a scheme over $k$. Assume*

(1) *$X$ is quasi-compact, and*
(2) *the connected components of $X_{\overline{k}}$ are open.*

*Then*

(a) *$\pi_0(X_{\overline{k}})$ is finite, and*
(b) *the action of $Gal(\overline{k}/k)$ on $\pi_0(X_{\overline{k}})$ is continuous.*

*Moreover, assumptions (1) and (2) are satisfied when $X$ is of finite type over $k$.*

*Proof.* Since the connected components are open, cover $X_{\overline{k}}$ (Topology, Lemma 004T) and $X_{\overline{k}}$ is quasi-compact, we conclude that there are only finitely many of them. Thus (a) holds. By Lemma 0388 these connected components are each defined over a finite subextension of $\overline{k}/k$ and we get (b). If $X$ is of finite type over $k$, then $X_{\overline{k}}$ is of finite type over $\overline{k}$ (Morphisms, Lemma 01T4). Hence $X_{\overline{k}}$ is a Noetherian scheme (Morphisms, Lemma 01T6). Thus $X_{\overline{k}}$ has finitely many irreducible components (Properties, Lemma 0BA8) and a fortiori finitely many connected components (which are therefore open). □